

Week 4 Assignment 2

Introduction to Wireless Communications

Submitted by: Syed Muhammad Ibrahim

Code explanation:

The Week 4 code simulates the Bit Error Rate (BER) performance of two digital modulation schemes, BPSK and QPSK, over two different channel models, AWGN and Rayleigh fading. Separate functions are written for each combination using their respective theoretical BER formulas. The Q-function is implemented using scipy's `erfc` function, and SNR is swept from 0 to 30 dB and converted from dB to linear scale for use in the formulas.

The results are plotted on a log-scale BER vs SNR graph showing all four curves alongside a target BER of $1e-3$. The simulation confirms that BPSK and QPSK perform identically over AWGN, both achieving the target BER at around 6.8 dB. However over a Rayleigh fading channel, significantly more SNR is required — roughly 17 to 20 dB extra — because random fading caused by multipath propagation with no line of sight severely degrades signal reliability compared to the ideal AWGN case.